

SIDDHARTH CHAINI

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EDUCATION

University of Delaware

Newark, DE, USA

Ph.D. in Physics and Astronomy (*GPA: 3.96/4.0*)

Sep. 2022 – Apr. 2027 (expected)

NASA FINESST Fellow & UD Doctoral Fellow for Excellence

Advisor: Federica B. Bianco

Dissertation: “*Discovering Structure and Surprise in Time-Domain Astrophysics*”

Data-driven discovery pipelines for next-generation surveys focusing on out-of-distribution anomaly detection of transients & variables, data-driven modelling of transients via reconstruction of sparse multisurvey & multifilter light curves, and physical characterization of supernovae spectra via data-driven methods.

Indian Institute of Science Education and Research (IISER) Bhopal

Bhopal, India

BS-MS (Dual Degree) in Physics (*CPI: 9.04/10*)

Aug. 2017 – May 2022

PUBLICATIONS



* marks equal contribution; author lists beyond three names are shortened to “et al. (incl. S. Chaini)”.

Lead & Major-Contribution Publications

- [13] **S. Chaini**, W. F. Fortino, S. Daniels, et al., “Data-Driven Representations of Supernovae Spectra: From Physical Features to Evolution.” *In preparation*.
- [12] N. Earl, **S. Chaini**, K. D. French, et al., “NAPTIME: A Neural-Process Framework for Rubin Alert Classification,” *ApJ*, Jul. 2026. arXiv: [2607.19236](#). *In press*.
- [11] **S. Chaini**, F. B. Bianco, and A. Mahabal, “Probabilistic Data-Driven Modelling of Astrophysical Transients: The Neural Process Family for Ultrafast and Class-Agnostic Light Curve Reconstruction with NightLANP,” May 2026. arXiv: [2605.27527](#). *Under review, NeurIPS main track*.
Also accepted as a workshop paper at ICML 2026, AI for Physics [\[OpenReview\]](#).
- [10] **S. Chaini**, F. B. Bianco, and A. Mahabal, “In Search of the Unknown Unknowns: A Multi-Metric Distance Ensemble for Out of Distribution Anomaly Detection in Astronomical Surveys,” in *NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences (ML4PS)*, Oct. 2025. arXiv: [2510.23702](#).
- [9] B. Domínguez*, **S. Chaini***, K. Mrakovčić*, et al., “Validation of the Bond et al. (2010) SDSS-derived kinematic models for the Milky Way’s disk and halo stars with Gaia Data Release 3 proper motion and radial velocity data,” *A&A*, vol. 688, A132, Aug. 2024, Art. no. A132. DOI: [10.1051/0004-6361/202449653](#). arXiv: [2406.03541](#).
- [8] **S. Chaini**, A. Mahabal, A. Kembhavi, et al., “Light curve classification with DistClassiPy: A new distance-based classifier,” *Astronomy and Computing*, vol. 48, p. 100850, Jul. 2024, Art. no. 100850. DOI: [10.1016/j.ascom.2024.100850](#). arXiv: [2403.12120](#).
- [7] **S. Chaini**, A. Bagul, A. Deshpande, et al., “Photometric identification of compact galaxies, stars, and quasars using multiple neural networks,” *MNRAS*, vol. 518, no. 2, pp. 3123–3136, Jan. 2023. DOI: [10.1093/mnras/stac3336](#). arXiv: [2211.08388](#).
- [6] **S. Chaini** and S. S. Kumar, “Astronomical Classification of Light Curves with an Ensemble of Gated Recurrent Units,” Jun. 2020. arXiv: [2006.12333](#).

Other Co-Authored Publications

- [5] C. R. Bom, A. Mahabal, F. Bianco, et al. (incl. **S. Chaini**), “Toward decision-aware AI for LSST-scale time-domain astronomy,” Jun. 2026. arXiv: [2606.05285](#). *Under review, Nature Astronomy*.
Rubin LSST Transients and Variable Stars Science Collaboration.
- [4] E. J. Honaker, J. E. Gizis, C. Aganze, et al. (incl. **S. Chaini**), “Searching for Ultracool Dwarfs in Early LSST Data Products,” ApJ, vol. 1004, no. 2, p. 214, Jun. 2026, Art. no. 214. DOI: [10.3847/1538-4357/ae6ce9](#). arXiv: [2604.08387](#).
- [3] C. L. Ransome, B. M. Subrayan, D. J. Sand, et al. (incl. **S. Chaini**), “First Results from the LSST Shadow Survey: The Restless Luminous Blue Variable AT2017des in the Virgo-Cluster Galaxy, NGC4532,” ApJL, Jun. 2026. arXiv: [2606.23784](#). *Under review, ApJL*.
The Shadow Collaboration.
- [2] A. Crawford, F. Bianco, T. A. Pritchard, et al. (incl. **S. Chaini**), “Quoth the Maven: Testing a Foundation AI Model on the Classification of Single- versus Double-peaked Stripped-envelope Supernova Light Curves,” 2026. *In preparation*.
- [1] D. J. Sand, B. Subrayan, C. L. Ransome, et al. (incl. **S. Chaini**), “Nearby Supernova Science in the Era of High Cadence, Deep Time Domain Surveys,” 2026. *In preparation*.
The Shadow Collaboration.

Software

NightLANP — [github.com/sidchaini/NightLANP](#) (2026).

keras-neural-processes — [github.com/sidchaini/keras-neural-processes](#) (2024).

DistClassiPy — [github.com/sidchaini/DistClassiPy](#) (2024); ascl:2403.002.

TECHNICAL SKILLS

Surveys & data	Rubin/LSST (DP0–2; Alerts: ALeRCE, Fink), DECam, ZTF, TESS, SDSS, Gaia
Languages	Python, C, C++, Java, Wolfram Language, SQL, HTML, Bash, Assembly, L ^A T _E X
Libraries	PyTorch, TensorFlow/Keras, scikit-learn, Astropy, NumPy, pandas, Matplotlib, Selenium
Tools	Git, Jupyter, VS Code, Mathematica, SAOImage DS9, Aperture Photometry Tool

FELLOWSHIPS, GRANTS AND AWARDS

- **NASA FINESST** — Grant 80NSSC25K0312; **US \$100,000** 2025 – 2027
Proposal written as Future Investigator; 24/456 Astrophysics proposals selected (~5%)
- **NSF Research Grant** — Grant 2511639; **US \$395,000** 2025 – 2028
Co-author on the funded proposal (PI: F. B. Bianco); graduate student researcher
- **Doctoral Fellowship for Excellence** — University of Delaware; **US \$39,000** 2025 – 2026
2 awarded per year in Dept. of Physics & Astronomy
- LINCC Frameworks Community Event Support, *LSST Discovery Alliance* 2026
- [Data Science Institute Fellow](#), *University of Delaware* 2023 – 2027
- Vatican Observatory Scholarship, *18th Summer School* – 25 selected from ~200 applicants, 20 countries 2023
- [INSPIRE Scholarship](#), *DST, Government of India* – Top 1% of ~1.2 million applicants nationally 2017 – 2022

OBSERVING EXPERIENCE

- **Dark Energy Camera (DECam)**, Cerro Tololo Inter-American Observatory – Remote observer, DECam Shadow project 2 nights

ACADEMIC LEADERSHIP & SERVICE

- **Elected Member**, LSST ISSC Executive Council Jun. 2026 – Present
Informatics and Statistics Science Collaboration — oversight of working groups, committee appointments and collaboration policies
- **Co-Chair**, LSST TVS Anomalies Subgroup Sep. 2024 – Present
Rubin LSST Transients and Variable Stars Science Collaboration — organize meetings and coordinate subgroup activities
- **Organizing Committee**, LSST is here! DP2 Hackathon Aug. 2026
Co-organized a 2-day hackathon on Rubin Observatory's LSST Data Preview 2 (DP2) data and tools
- **Instructor**, Rubin TVS Software Taskforce – Multiple collaboration-wide software workshops Aug. 2026
- **Reviewer**, workshop programme committees – ICML Workshop on AI for Physics; NeurIPS Workshop on Representations for Physical Sciences 2026
- **Elected Head**, Student Research Group, IISER Bhopal Astronomy Club Aug. 2020 – Aug. 2021
In charge of group activities and data analysis — image reduction and CCD photometry

TALKS, PANELS & POSTERS

Invited:

- Quasar Bazaar Hackweek, SISSA, Trieste, Italy ([Talk](#)) Feb. 2026
Distance Multi-Metric Anomaly Detection for LSST Transients & Variables
- Astronomy Seminar, National Centre for Radio Astrophysics, Pune, India ([Talk](#)) Jan. 2026
Transient hunting in the era of large-sky surveys with Neural Processes
- AI/ML Applications in Astronomy and Astrophysics, IUCAA, Pune, India (Panelist) Jan. 2025
Pitfalls of using ChatGPT
- LSST TVS Colloquium ([Talk](#)) Oct. 2024
Distance Metrics for Machine Learning in Time-Domain Astronomy with DistClassiPy

Contributed:

- Physics in Supernovae Spectral Manifolds
European Astronomical Society 2026 ([poster](#)).
- Probabilistic Light Curve Modelling with NightLANP
Rubin Community Workshop 2026 ([talk](#)), Extragalactic Transient Universe ([talk](#)), European Astronomical Society 2026 ([talk](#)), APS Global Physics Summit 2026 ([talk](#)), TDABench 2026 ([talk](#)), LSST@Europe 7 ([talk](#)), IAU Symposium 406 ([poster](#)).
- Distance Multi-Metric Anomaly Detection
Rubin Community Workshop 2025 ([talk](#)), NeurIPS: ML and the Physical Sciences 2025 ([poster](#)), Shaping the Future of Time-Domain Astronomy with LSST ([talk](#)).
- Light Curve Classification with DistClassiPy
AI/ML Applications in Astronomy and Astrophysics ([tutorial](#)), LSST@Europe 6 ([talk](#)), Rubin Community Workshop 2024 ([talk](#)), Rare Gems in Big Data 2024 ([poster](#)), APS Mid-Atlantic Section 2023 ([poster](#)).

MENTORING

- Co-mentored three Lincoln University undergraduates May 2026 – Aug. 2026
Abdourahim Sylla, Agustina Lorda and Adeoluwa Afolabi
- Co-mentored UD undergraduate NASA Space Grant Fellow Jul. 2023 – Aug. 2024
Ibrahim Wilson

TEACHING EXPERIENCE

- **Lesson Instructor**, Rubin LSST TVS Science Collaboration Jul. 2024 – Present
Intermediate Python for Astronomical Software Development; Research Software Engineering Mini Course
- **Teaching Assistant**, University of Delaware Sep. 2022 – May. 2025
PHYS 207/227: Fundamentals of Physics; PHYS 208: Electromagnetism; SCEN 101: Physical Science
- **Teaching Assistant**, IISER Bhopal Jan. 2019 – May 2019
ECS 102: Introduction to Programming

OUTREACH

- **DP2 Hackathon**, University of Delaware — co-organized a hackathon targeted primarily at ECRs in the Mid-Atlantic region to share and work on Rubin DP2 science together.
- **Rubin First Look**, University of Delaware — co-led a public viewing party for Rubin Observatory’s first images, and presented LSST’s goals and science to the public.
- **Astronomy Club**, IISER Bhopal — organized telescope viewing nights and computational-astronomy programming tutorials for the campus community.
- **Outreach talks**, tutoring centres in Thane, India — science talks to students in grades 8–10 to spark interest in astrophysics.
- **Media coverage** — *UDaily*: “NASA’s next generation of scientists” (Oct. 2025), “Opening a new window into the Universe” (Jun. 2025); *Franciscan Media*: “Vatican hosts astronomers of tomorrow” (Jun. 2023, also on YouTube).

REFERENCES

- **Federica B. Bianco** (University of Delaware) — *PhD advisor* fbianco@udel.edu
- **Ashish Mahabal** (Caltech) — *MS thesis co-advisor* aam@astro.caltech.edu
- **Željko Ivezić** (Rubin Observatory/UW) — *mentor* ivezic@uw.edu